CHAPTER 2

THE BRAIN

NUMEROUS ANALOGIES HAVE been proposed to help explain, clarify, and contextualize both the arrival and the significance of artificial intelligence. Anthropologists liken it to fire or electricity. Generals and diplomats postulate a resemblance to atomic power or to an unstoppable, unconquerable human force of will like Otto von Bismarck. Astronomers describe it as akin to the arrival of an asteroid—a distant and low-probability prediction around which humans might organize a planetary defense—

or the discovery of alien life. Economists analogize AI to bureaucracies and markets, while leaders of state and society compare it with the arrival of the printing press or the corporation—the latter growing to wield a will of its own, and in an early instance taking over the Indian subcontinent before the world understood its incompatibility with and potential domination of existing structures of power.[1]

Today, our own view is otherwise: No innovation, no matter how profound, can come as close to the original inspiration and (we believe) now-temporary destination for our quest to build intelligence: namely, intelligence that is greater than any human's on the planet.[2]

There are two ways of thinking about our present circumstances. The first is a projection of the familiar. To date, humanity's most transformative technologies have enhanced or amplified human bodily function. The wheel reduced the exhaustion of increased

mobility, while engines of various types relieved the agony of torn muscle. X-rays, magnification, and the light bulb stretched the limits of observable reality beyond natural eyesight alone, just as the telephone amplified our voice in ways our throats could not. All dimensions of human function have, in some way, been inorganically augmented, sharpened, or strengthened by machines of our creation. Is AI, then, just another extension of human faculties?

The second way of thinking is to suggest that, this time, things are different—that there are unique aspects to AI that are not augmentations of human abilities. By engineering in a matter of decades a counterpart to what evolution produced over millennia—that is, the brain—we have found ourselves tackling the last organ remaining for inorganic replication or reinvention.

SPEED

In the previous chapter, we pointed to the similarity between the training of an AI machine and the training of the mind of an advanced student of philosophy. That example can be extended more broadly. Put plainly, the formation of mechanical intelligence can be seen as parallel to the process by which the human brain matures biologically from adolescence into adulthood.

Students during the course of their secondary education learn the fundamentals of core subjects, building up their basic view of the world. That view may not be particularly advanced—or always correct—but the same is also true of a machine. Machines, like humans, learn by absorbing information and transforming it into theory for subsequent practice. When machines learn, an algorithm ingests vast amounts of data—scraped from sources on the open internet or provided more specifically by other, private sources—and collapses the results into a condensed and compressed mapping of concepts for future use. Just as humans' biological mechanisms map sensory input onto neural "weights" that connect the network of the brain's processing units, machines similarly require a gradual

strengthening of their own computational weights.

Neural networks, like (some) high-school students, can be lazy. During the early stages of training, AI will do the bare minimum. Memorizing answers rather than actually learning, a model faced with "2 + 2" might initially encode the answer "4" without having mastered the underlying principle of addition. But rapidly, across a certain threshold, this approach will break down, forcing the machine to abstract upward—as humans do—to more universal axioms of knowledge.

This is what principally distinguishes AI from ordinary computers: Its mapping of the world is not programmed, but learned. In traditional software programming, a human-created algorithm instructs a machine in how to transform a set of inputs into a set of outputs. In machine learning, by contrast, human-created algorithms tell the machine only how to improve *itself*, allowing the machine to design its own mappings for the input-to-output transformation. As the machine "learns" from countless prior trials, failures, and adjustments, it upgrades its algorithms, iteratively redesigning its internal mappings of the patterns and connections it "sees" in the data.

Periodically, human trainers will give the machine feedback on the accuracy and quality of its outputs. The machine internalizes their corrections by means of "back-propagation," a technique that allows the effects of the trainers' changes to ripple backward through the mathematical relationships that the machine has already created, thereby improving the overall model.

For any given model, however, humans provide feedback on only a small subset of possible inputs and outputs. After the model performs at a certain level in a number of training tests, its developers trust that its established mappings will generate a safe and accurate response to all inputs, even unexpected ones, with a high probability of success.

In each of these ways, AI is already expanding, and will further expand, the realm of human knowledge. But it is doing so—and we are accepting the resultant knowledge as true—by processes that we do not fully understand.

Where a typical student graduates from high school in four years, an AI model today can easily finish learning the same amount of knowledge, and dramatically more, in four days. And thus, speed has proven itself to be the first in a handful of core attributes that distinguish AI from our human form and mental capabilities.

Despite having highly advanced parallelism—that is, the ability to process simultaneously different kinds of stimuli—the human brain is a slow processor of information, limited by the speed at which our biological circuits work. If a human brain's circuits were analyzed with the same performance metrics as computers—by "clock rate" or processing speed—the average AI supercomputer is already 120 million times faster than the processing rate of the human brain.

True, speed is not a strong indicator of intelligence; very dumb humans can think quickly. Nevertheless, a faster pace of processing provides two benefits as compared with the human brain: the ingestion of vastly more information and the service of many more simultaneous requests. Much of the human brain typically remains on autopilot—passively serving internal needs in guiding the beating of our heart and the movement of our limbs, and intervening to adjust only when the autopilot proves faulty. By contrast, the speed of which AI is capable allows for the programmatic emergence of great prowess, which then enables the achievement of higher, more difficult, and grander problems than those currently solvable by the human brain.

Once both the human and the machine have completed their intellectual training, both are now theoretically capable of "thinking" or, in the equivalent technical term, "inference." In the course of an interview, an argument, or a date, a student-turned-graduate draws upon his or her education and experience. We do this not by regurgitating exact formulas, individual facts, and precise numbers from memory but by consulting a thinner layer of contemplation of and reflection on what we have learned. The human brain was never meant to memorize information for perfect recall; nor are most brains capable of doing so. What should remain instead, after innumerable lessons, essays, and exams, is a grasp of the deeper and longer-lasting concepts that those same educational tools of

instruction are meant to reveal: the wonder of astronomy, the tragedy of ambition, the necessity (or not) of revolution.

The same is true of AI. When a model emerges from the completion of its training run, it no longer requires access to the original data it was trained on. Rather, it is left only with a rough guiding intuition, assembled from the knowledge it has received, for answering questions, challenging reasoning, and making predictions. Just as humans don't haul libraries of material around with them, an AI model similarly infers rather than recalls. The difference, then, is that superior speed facilitates this inference across a broader, deeper array of learned information than a human could ever hope to attain.

To do this, even to answer a simple question, an AI model may perform billions of complex technical operations. Whereas a traditional computer simply retrieves specific information stored in its memory—since it is unable to arrive at the sorts of conclusions that don't already exist there—AI launches computing in the direction of the human brain. Just as humans *learn* in order to *think*, machines *train* in order to *infer*. The second cannot come without the first.

The first phase—for both humans and machines—is the more intensive process, in both the amount of time spent and the number of resources required. A postdoctoral student may have spent two decades or more building the capability to compose—in two days—a thoughtful essay on a given subject. Similarly, training the largest AI models may take months, but the resultant inferencing can consume mere fractions of a second.

Today's AI systems already give apparently cogent and considered answers in response to human queries. In their latest and future iterations, they will operate comprehensively, crossing multiple domains of knowledge with an agility that exceeds the capacity of any human or any group of humans. For AIs, scale—in the sense of size—enables speed; as we have just seen, the larger and more thoroughly trained the machine, the faster and more exhaustive its results. What is more, by recognizing patterns in the data that elude the inquiring human operator, AI systems will be equipped to distill traditional expressions of knowledge into original responses and, out of enormous amounts of data, forge new

conceptual truths.

Which raises a question, or rather more than one question.

OPACITY

How do we know what we know about the workings of our universe? And how do we know that what we know is true?

In most areas of knowledge, ever since the advent of the scientific method, with its insistence on experiment as the criterion of proof, any information that is not supported by evidence has been regarded as incomplete and untrustworthy. Only transparency, reproducibility, and logical validation confer legitimacy on a claim of truth. Under the influence of this framework, recent centuries have yielded a huge expansion in human knowledge, human understanding, and human productivity—culminating with the invention of the computer and machine learning.

Today, however, in the age of AI, we face a new and peculiarly daunting challenge: information *without* explanation. Already, AI's responses—which, as noted above, can take the form of highly articulate and coherent descriptions of complex concepts—arrive instantaneously. The machines' outputs are delivered bare and unqualified, with no apparent bias or motive—but also unaccompanied by any citation of sources or other justifications. And yet, despite this lack of a rationale for any given answer, early AI systems have already engendered incredible levels of human confidence in, and reliance upon, their otherwise unexplained and seemingly oracular pronouncements. As they advance, these new "brains" could appear to be not only authoritative but infallible.

Although human feedback helps an AI machine refine its internal algorithms, the machine holds primary responsibility for detecting patterns in, and assigning weights to, the data on which it is trained. Nor, once a model is trained, does it publish the internal mathematical schema that it has concocted. As a result, the representations of reality that the machine generates remain largely

opaque, even to its inventors. Today, humans attempt to assure themselves of the integrity of these machine models mainly by examining outputs alone. The internal workings remain largely impenetrable—hence the reference to some AI systems as "black boxes." Although some researchers are attempting to reverse-engineer the outputs of these complex models into familiar algorithms, it is not yet clear whether they will succeed.

In brief, models trained via machine learning allow humans to *know* new things (the models' outputs) but not to *understand* how the discoveries were made (the models' internal processes). This separates human knowledge from human understanding in a way that would have been foreign to any other age of humanity. Human apperception in the modern sense has developed from the intuitions and outcomes that follow from conscious subjective experience, individual examination of logic, and the ability to reproduce the results. These methods of knowledge derive in turn from a quintessentially humanist impulse: "If I can't do it, then I can't understand it; if I can't understand it, then I can't know it to be true."

In the framework that emerged in the Age of Enlightenment, these core elements—individual human capacity, subjective comprehension, and objective truth—all moved in tandem. By contrast, the truths produced by AI are manufactured by processes that humans cannot replicate. Machine reasoning, which does not proceed via human methods, is beyond human subjective experience and outside the capacity of humans, who cannot even fully represent the machines' internal processes.

These facts, by Enlightenment reasoning, would preclude the acceptance of machine outputs as true. And yet we—at least, the millions of humans who have begun to interface with early AI systems—have already accepted the veracity of the vast majority of their outputs.[3] Granted, some advanced users may indeed actually comprehend the meta-process of machine learning; for most, however, human confidence in the objective truth of the machines' outputs must rest in a type of faith that expresses itself as a wishful

belief in the machines' logic and their developers' authority.

In itself, the emergence of such belief as an accepted method in the pursuit of objective truth marks a major transformation in modern human thought. For even if AI models do not "understand" the world in the human sense—because machines emphatically do not experience consciousness or subjectivity—their objective capacity to reach new and accurate conclusions about our world by nonhuman methods not only disrupts our reliance on the scientific method as it has been pursued continuously for five centuries but also challenges the human claim to an exclusive or unique grasp of reality.

What can this mean? Will the age of AI not only fail to propel humanity forward but instead catalyze a return to a premodern acceptance of unexplained authority? In short, are we, might we be, on the precipice of a great reversal in human cognition—a dark enlightenment?

DIVERSITY

Different entities measure time along different scales. On a geological timescale, the entirety of human existence would appear as a tiny dash at the tail end of Earth's 4.5-billion-year span. If we humans were progressing at geological speed, we would perceive only stasis. Instead, as an impatient and self-important species, we have defined our own pace of evolution. Whereas an "age" in geological time is measured in the thousands of years, human time measures an "age" as a century or two.

As for an artificial or technological timescale, AI would likely operate at its own distinctive measure. The whole history of artificial intelligence spans no more than 70 years. Just as humans generally consider the many hundreds of millions of years before the Cambrian explosion to have been an impossibly long blank slate before a sudden outburst of animal life and evolutionary progress, AI would probably characterize the six decades of 1950 to 2010 as a similarly

slow, murky period of nothingness, illuminated by glimmers of life only at the very end.

Human generations, to judge by society and biology, last some 25 years. AI, by contrast, moves with inhuman speed; its generations are much shorter, with leaps happening in perhaps a tenth of that time. We should therefore anticipate that what may feel in human time like a revolution will appear in technological time as a mere evolution. Newer AI models—with only months in between—can respond to ever more open-ended prompts, make more choices in order to reach a given goal, and act across increasing numbers of modalities.

Thus, the age of AI—in human time, perhaps a hundred years—might more accurately be labeled the *ages* of AIs, and, according to the technological timescale of AI itself, might be said to encompass many hundreds of generations.

The rapidity of AI's evolution is a multifaceted—and largely unappreciated—challenge. Humanity has never before had to deal with, or to prepare for, such a temporal compression. The sheer speed of the change practically ensures cultural and psychological disorientation. As the effects of the new technologies on daily life layer and compound, they will complicate efforts to pinpoint any one application as either a source of crisis or, to the contrary, a comforting harbinger of progress.

Untangling these overlapping issues in the real world will become ever more difficult as a multiplicity of AIs brings with it a multiplicity of inscrutable effects. Moreover, as AI grows more powerful, the future will likely bring a significant evolution and diversification. New infrastructures and techniques for machine learning, so long as they are unconstrained, will spawn generations of AI with increasing diversity, breadth, capacity, and complexity. Just as electricity powers more than light bulbs, AI will be put to any number of uses. And just as there are many ways to generate an electrical charge—friction, conduction, induction—we may anticipate the discovery of multiple ways to create AI.

For instance, the infrastructure that has made recent AI advancements possible is known as a "transformer." It allows the

machine to consider, for instance, the connections among multiple words at once. In lay-speak: where previous structures read words one at a time, capturing only the connection between word 1 and word 2, then separately the connection between word 2 and word 3, a transformer allows the model to capture, all at once, both a whole sentence and every connection between every word in the sentence. Creating and utilizing mathematical representations of all these connections, the AI predicts the best response.

The capabilities of transformers were not anticipated, and their highly generalizable successes occurred almost incidentally.[4] And transformers are not necessarily the only foundational infrastructure that can yield unexpected capabilities. As more fruitful avenues of research emerge, AI outputs will improve quickly, multiplying along different lines of physical and mathematical logic, at lower expense and faster speed.

In its evolutionary speed and diversification, AI's development will thus indeed be akin to the Cambrian explosion: the emergence of a wide variety of different forms of life within a single, highly compressed period of time relative to the preceding epoch. If this conjecture is correct, machine intelligences will branch outward into a rapidly evolving genus, or even a family, of many different AIs operating on many different forms of logic. AIs may thus offer the most striking instance of the diversity that can come from minor changes iterated repeatedly across the board: a digital echo of the organic world. As Darwin writes, "from so simple a beginning, endless forms most beautiful."[5]

SCALE AND RESOLUTION

The Age of Reason may have brought humanity to the edge of how humans understand our world. Einsteinian physics and the formulations of quantum mechanics were the beginnings of a still incomplete venture into uncharted territory: worlds potentially with their own rules of knowledge, apprehensible not by applied

perception but only by theoretical ideation. Quantum mechanics describes the world at the micro-scale, where, as the Harvard physicist Greg Kestin puts it, "Nothing is predictable and objects don't have precise positions *until* they are observed," and general relativity describes the world at a cosmic scale, where everything is predictable, "*whether or not*" observed.[6] Neither theory has failed, but both cannot be true, and "No experiment has been able to show which—if either—of the two theories" reigns supreme.

Ironically, this uncertainty underlies the modern world. Quantum physics enabled—among other revolutions—the computing revolution. AI is and will be very much the same. Already, it produces insights and transforms reality by mechanisms that we don't fully understand. And soon enough it will be occupied with a science that is even less comprehensible to human understanding.

After three hundred years, the Age of Reason, despite its manifold successes, has stalled—as is evidenced by our manifest inability to make further progress on the unification of physics. Our present strain and struggle in these days of unassisted human science, more than a century after the conception of the core theories underlying both the cosmic and quantum worlds, is but one sign that humans may be nearing some biological limit of intelligence.

Because of its unique methods of inquiry and learning, AI will be capable of inhuman achievements in terms both of size ("scale") and of precision ("resolution"), thereby activating fundamental changes different from any other human invention or from the human species itself. Nevertheless, could AI achieve a reconciliation between the two ends of human reality, inducing a revolution in perception by methods that until now have been completely foreign to human experience?

The physical scale of our human brain is dictated by our anatomy. Human brains must fit inside human skulls, and infant human skulls must, generally, fit through the female birth canal. Any smaller, and such humans may be at a cognitive disadvantage; any larger, and those babies—or their mothers—may not survive childbirth. Other

physiological limitations, too, enforce a ceiling of constraint—practical limitations on the weight of the brain, for example. Barring Cesarean sections or, later, artificial wombs, this means that humans have reached an evolutionary equilibrium.

For AI, today's models have capabilities that were not anticipated at their creation. The "scaling laws" (like, in an old-fashioned example, the laws governing the relationship between the length of an object and its area) that have applied so far to AI seem to be holding true, but we do not know what precisely will be enabled by models with an exponentially increasing number of parameters because we have not uncovered a scientific reason why certain capabilities emerge at a particular degree of power and complexity.

Brain size relative to body size does not correlate clearly with intelligence in the animal kingdom—dolphins, elephants, and some whales all have brains proportionately larger than the human brain. But early science does suggest that scale plays *some* role, which we do not yet understand.

Given our hard biological constraints, humans are unlikely to test their own brains' "scaling laws." But AI enters the world with no prefixed physical size. It is not bound to any physical carrier of discernible scale. Chips and data centers—the physical hosts of AI models—can be clustered and connected without as yet observable limit. In other words, the scaling laws will surely be tested for AIs as they never have been for humans. And as they are tested, scale—which has constrained the range of human understanding throughout the history of scientific thought—may turn out to be the primary differentiator between human brains and AI models.

One of the key side effects of scale will be resolution. Humans have long desired to extend the range of what we can observe to both the very tiny and the very far. The microscope and the telescope are quintessential tools of human observation. Less appreciated is the humble pen. Writing, invented four thousand years ago, remains an outstanding tool for the codification and transmission of complexity. That includes mathematics, perhaps the purest and most universal of human languages, and enough in itself to facilitate the transfer of abstruse ideas and collaboration on

technological projects. On a per-byte basis, language in all its multifarious and beautiful forms is unusually dense—among the most efficient data structures that have been invented.

Even after we enlarge or compress reality to produce observable information, humans must perform a second step: We must abstract away from raw information in order to make that information useful. AIs today do the same. And they do so using tools that mirror our own: binary strings of zeroes and ones, the translation of documented human experience into the language of computers. Like writing, these strings of language appear rudimentary in retrospect. Yet they have enabled digital representations of both sight and sound: the human senses of highest bandwidth.

Despite these similarities, AIs will diverge from humans. As the scale of an AI increases, it will be able to process larger volumes of information simultaneously and produce analysis that is useful (at least to itself) without unnecessarily sacrificing granularity. The scale of the data on which an AI is trained, combined with the complexity of its network and the density of the symbols on which it operates, appears to yield unprecedented resolution in its processes of learning and inference, and ultimately in its outputs. It is an elegant inversion that early AI, trained on the text of the internet—the universal library of humanity and the decentralized web of our compressed experiences—is likely to unlock entirely new knowledge for humanity about ourselves, in senses both cosmic and microscopic.

THE ANIMAL KINGDOM

At our current moment, some will undoubtedly decry any close comparison of AI with the human brain. To a human, concepts are rich with meaning and vessels for deep expressions of joy or sorrow. In contrast, machine understanding can seem fake. Although it may soon be able to produce eloquent works on common themes of humanity that eclipse even the best human authors, an AI does not

search for or grasp attendant meaning. Thus, exploring the human condition by reverse-engineering the language of human writers seems, at best, a coldly superficial mastery of linguistic probabilities. That some random complex engine can take language—a gift so organically human—and use it as a hyperefficient means to ingest information can be as upsetting as it is confusing.

But our own biological circuitry may be just as mechanical as silicon circuitry, and the processes by which our human brains work do not appear to be any more special than the way machines already operate. We are far from possessing a complete theory of neuroscience, but we do know that our brains, like AI models, are largely powered by predictive processing. That is, when listening or reading, our human brains contain a neurological predictor that lends assistance by anticipating the next word in a verbal string. Without such mechanisms, we would be physically and psychically exhausted by the sheer amount of effort required for even the simplest of cognitive tasks.

And these predictive powers, like AI's, have provided the foundation for human mastery of our world. All the most advanced representations of human knowledge have been built atop language and symbology, allowing us at once to reproduce works of complex engineering and to communicate the anguish of heartbreak.

AI has been likened to the lifelong prisoners in Plato's cave who, without having known anything else, believe that the shadows cast on their walls are the full extent of reality.[7] Just so, humans assume that machines are devoid of context and that a machine's perception is limited by the bounds of the material on which it trains, without any capability to probe or infer further.

Hubris, perhaps, is what prevents us from seeing the similarities between organic and inorganic brains, and from admitting the possibility of the latter's capacities. Already, AI systems are showing evidence of perceiving a universe that exists beyond the confines of the dataset used to construct their custom slice of reality. AIs may be capable of receiving deeper meaning, after all—even if they are not seeking it. All it would take is for one enterprising prisoner to

theorize, by chance, that the shadows on the wall may be representations of a larger world with higher dimensionality.

If such a breakthrough were to happen, it should not be altogether surprising, considering the speed, complexity, diversity, scale, and resolution of these new intelligences. Nevertheless, it may be intensely disruptive. The appearance of knowledge—especially of the physical world—that is unique to an AI and not previously possessed by a human would force a reconsideration of the relative status of the human mind. Humans placing our own brains on new, more continuous spectrums of intelligence would revolutionize our perceptions, self-perceptions, and behaviors.

This is not to say that AIs will exclusively and immediately exceed all human beings in intelligence. But there will be phases in the evolution of AI when mechanical intelligence may feel eerily similar to the intelligence of the animals. A disorienting debate will surely ensue as we attempt to reorganize a long-perceived hierarchy of beings, measuring downward from humans, to animals, to machines. Human intelligence will be urgently pressed to confront the reality that it is no longer the sole, or even the superior, model of intelligence.

AI models are already being used to assist humans in deciphering—and responding to—the communications of animals. Early experiments in decoding high-pitched clicks and trumpeting calls are catalyzing revisions to our long-standing bias that humans are special or distinct from other animal species. The ability of humans and animals to commune with each other directly—no longer, in our case, through body language or facial expressions alone—may stimulate the reeducation needed to prepare us for what is to come with AI.

To be sure, animal-human-machine communication would yield a complicated trilateral negotiation. Our world would be populated by beings—new and old—fighting to secure a new position or to retain an existing one. Machines may contend that the truest method of classification is to group today's humans together with other animals, since both are carbon systems emergent of evolution and different from silicon systems emergent of engineering. It might be

difficult for machines to be confident (in a numerical sense) that humans are superior to other animals by any standard of measurement. If this were not the makings of a tragedy, it would be the stuff of comedy.

How intelligent must any intelligence—biological or mechanical—be in order to be recognized as our equals? Animals of lesser but still high intelligence may, upon articulating and negotiating their terms of existence, cause us to reevaluate our treatment of them. They might deserve, and convince us that they deserve, a unique and hitherto unacknowledged right of existence or independence.

Precisely as is now often advocated for animals, some humans may be moved in turn to advocate a coterminous standard of treatment for humans and AIs. Indeed, humanity should not embrace weaker morals even if logic were to lead us into a delicate position. On the other hand, one must be aware that this phase could be but a brief and transitory step before pity gives way to panic.

A TWOFOLD PARADOX

We are confident that AIs will outstrip the human brain in speed, diversity, scale, and resolution, reorganizing the hierarchy of intelligence that humans have thus far constructed. The extent of our potential disorientation and perceived inferiority from this change may rest on a relatively small detail: whether AI's structures continue to resemble the structures of the human brain.

Some AI researchers believe that approximating the human brain is the best path forward for the development of machine intelligence.[8] Here it suffices to note that, after all, the human brain is "the only existing proof" that such an intelligence is even possible.[9] But more likely seems a mixture of AIs and elements thereof, with some additional innovations and structures inspired by the brain and others of a different design.

In human brains, deep abstract thought and creativity seem to

require the use of neural systems beyond those needed for ordinary functioning. This science is still nascent, but it is possible that AIs will similarly need layers and add-ons to achieve increasingly advanced tasks of higher reasoning.

True, if AI's development were to continue to reflect, either by design or by serendipity, some approximation of the human brain, humans could indeed theoretically come to see their own excellence and meaning mirrored and extended in the achievements of machines. But if we hope to construct a machine that will vastly outperform the capacities of the human brain, is not divergence from the original blueprint an eventual necessity? Airplanes were inspired by birds, but not engineered to emulate them—and thus modern jet aircraft outperform the most advanced biology ever to roam the skies. Will we have any reason to believe that rebuilding the source of all invention from scratch will be any different?

More likely, the architects of AI will hold the human being to be our guide as well as our cautionary tale, its design scrutinized for both its functions and its flaws. The human brain, therefore, becomes not the goal, and not a blueprint, but a midpoint and an inspiration toward something greater.

In any other realm of human endeavor, to have a clearer conception of an intermediate design than of the final objective might well cast doubt on the feasibility of the entire effort. In our case, we face this particular paradox on a second level: We are trying to build something modeled on the brain—and superior to the brain—while still not fully understanding the brain itself. How does one exceed, even in design, what one does not understand to begin with? Without a precise understanding of either the means by which our "something" currently works or the ends by which it should work, our pursuit of something greater remains as formidable as it is also, quite wonderfully, thrilling.

We also face grand uncertainty as to the effects of such a development. If machine intelligence should continue to diverge from the example of the human mind, it will appear to us not as humanity's reflection but as its replacement. True, for a transitional period it might only amplify the range of activities today considered

“human abilities”; but after a certain point its own abilities could instead supplant the human variety, suggesting that our ideas of human excellence require total redefinition.

Future attitudes toward the very nature of human existence may hinge on this point. If our tools incorporate some or most of our intellectual and creative functions but do not mirror our own minds, could an emergent AI jeopardize deep-seated beliefs of humanity’s unique reflection of and special relationship with the divine? Or, in the alternative, might the apparently superior intelligence of machines with structures based on the human brain, combined with our intense reliance on them, lead some to believe that we humans are ourselves becoming, or merging with, the divine?